

# Social Media and EPSS Analysis

A plain-language account of the dataset, historical risk analysis, comparison with similar CVEs, external data sources, results, and verification.

This report follows the original social-media records through the temporal EPSS study, comparison with similar CVEs, and final interpretation. Telegram posts were excluded because their recorded dates represent message-crafting dates rather than original publication dates. For each CVE, the earliest post captured in this dataset defines the event, and historical EPSS is examined before and after it. Similar comparison CVEs are selected from the same dataset. Original post text supports basic quality checks and simple features, while LLM summaries are evaluated separately and do not determine the primary temporal results.

## Main finding

CVEs often move upward in EPSS percentile during the first seven days after their first captured post. The average difference from similar CVEs remains positive, but the median is close to zero, and the 30-day difference is not convincing. This is a short-term warning signal, not evidence that social media caused exploitation.

## 1. Purpose and Key Terms

This project asks whether social-media attention can add information about future vulnerability exploitation risk. A CVE is a public vulnerability identifier. EPSS estimates 30-day exploitation probability on a 0-to-1 scale. CVSS measures technical severity on a 0-to-10 scale. CISA KEV is an external list of vulnerabilities known to have been exploited.

A post-CVE row is one connection between one collected post and one CVE. The event date, written as  $t$ , is the earliest post date captured for a CVE in this dataset. The posted CVE is the vulnerability whose post defines the event. A comparison CVE is a different, similar vulnerability whose first captured post occurs after the posted CVE's study window. Statistical studies call this selection process matching, but this report uses comparison CVE.

An event window is the period around the post, such as seven days before and after it. Required pre-history means EPSS existed far enough before the post to calculate that window. CVE-specific text means the exact post text is connected to only one CVE, rather than being a general article or copied page linked to several CVEs.

The baseline anchor is the starting EPSS value on day 0, or the next available value on day 1 or day 3. Horizon means days after the event. EPSS percentile is the CVE's daily position compared with other scored CVEs. Mean is the average; median is the middle value. A small p-value supports evidence of a difference but does not prove that the post caused it.

## 2. Dataset Structure

Gemma, GPT, and Mistral files contain the same post and vulnerability records but different LLM-generated summaries. A row is a post-CVE link, so one CVE can appear more than once. Canonical CVEs remove the row suffix and count each vulnerability once.

| Dataset measure                      | Verified value |
|--------------------------------------|----------------|
| Rows in each original model file     | 9,218          |
| Columns in each file                 | 28             |
| Canonical CVEs in original files     | 5,692          |
| Rows used in this analysis           | 6,628          |
| Canonical CVEs used in this analysis | 4,187          |
| First-post events                    | 4,187          |

Rows count post-to-CVE connections, so one post mentioning several CVEs creates several rows. Canonical CVEs count each vulnerability once after removing the row suffix. First-post events retain the earliest captured date per canonical CVE.

| Model file                | Parsed rows | Independent CSV rows | Columns | Complete columns | Columns with missing values | Ragged rows |
|---------------------------|-------------|----------------------|---------|------------------|-----------------------------|-------------|
| gemma_combined_summ.csv   | 9218        | 9218                 | 28      | 21               | 7                           | 0           |
| gpt_combined_summ.csv     | 9218        | 9218                 | 28      | 21               | 7                           | 0           |
| mistral_combined_summ.csv | 9218        | 9218                 | 28      | 20               | 8                           | 0           |

The files were counted in two independent ways, and both found 9,218 rows and 28 columns. Complete columns have a value in every row. Columns with missing values counts fields containing at least one blank. A ragged row would contain too

many or too few columns; none were found.

### 3. Online Data Sources and Tools

- FIRST EPSS API: <https://api.first.org/data/v1/epss>. Queried by CVE and historical date for EPSS score and percentile.
- FIRST EPSS documentation: <https://www.first.org/epss/data.html>. Used to confirm daily data, historical coverage, and model-version changes.
- Historical archive: [https://epss.empiricalsecurity.com/epss\\_scores-YYYY-MM-DD.csv.gz](https://epss.empiricalsecurity.com/epss_scores-YYYY-MM-DD.csv.gz). Used in the coverage and feasibility audit.
- CISA KEV catalog: <https://www.cisa.gov/known-exploited-vulnerabilities-catalog> and its public JSON feed. Used for known-exploitation status and catalog-addition dates.
- Local tools: pandas for data preparation, SciPy for statistical tests, scikit-learn for models, Matplotlib for figures, and ReportLab for PDF generation.

Downloaded EPSS and KEV data were cached locally. The PDF was regenerated from those cached records and saved result tables, so a later website update could not silently change the reported values.

FIRST supplied historical EPSS scores for CVEs and dates. It did not supply the identities of the comparison CVEs. Both posted and comparison CVEs came from the social-media dataset, and the local selection algorithm decided which CVEs were similar enough to compare.

### 4. Important Attributes

| Field or field group       | Meaning and caution                                             |
|----------------------------|-----------------------------------------------------------------|
| cve                        | CVE plus row suffix; suffix removed before CVE-level analysis   |
| source                     | Source of the collected post; source coverage differs over time |
| date_posted / time_posted  | Recorded date and time; 00:00 is often a placeholder            |
| social_media_post          | Original text; may be reused or contain a full page             |
| epss_score / epss_status   | Exploitation probability and how the score entered the dataset  |
| cvss_score / submetrics    | Technical severity and the components used to calculate it      |
| occurrence_count           | Supplied count of appearances; may reflect collection behavior  |
| source_links / github_urls | External references and GitHub evidence                         |
| LLM summary fields         | Generated summaries of references, GitHub evidence, and CVSS    |

### 5. Source Profile

| Source           | Post-CVE rows | Unique CVEs | Mean EPSS | Median EPSS | Mean CVSS | Enriched EPSS rows (%) | First date | Last date  |
|------------------|---------------|-------------|-----------|-------------|-----------|------------------------|------------|------------|
| Mastodon         | 3136          | 2860        | 0.0085929 | 0.00043     | 8.8121    | 58.578                 | 2024-09-04 | 2025-05-20 |
| Reddit           | 1333          | 616         | 0.1495    | 0.00174     | 7.9374    | 5.7764                 | 2021-11-23 | 2025-06-01 |
| HackerNews       | 816           | 483         | 0.25256   | 0.00738     | 7.7561    | 4.6569                 | 2025-04-07 | 2025-05-28 |
| BleepingComputer | 755           | 276         | 0.19062   | 0.00228     | 8.1415    | 20.132                 | 2025-01-04 | 2025-05-30 |
| ExploitDB        | 588           | 362         | 0.1804    | 0.00146     | 7.7721    | 12.585                 | 2023-05-02 | 2025-05-29 |

A post-CVE row is one link between a collected post and one CVE. A post mentioning three CVEs can create three rows. Unique CVEs count each vulnerability once. Mean is the average and median is the middle value. Enriched EPSS rows show how often EPSS was added later. First and last dates show when each source appears in the dataset.

## 6. Figures

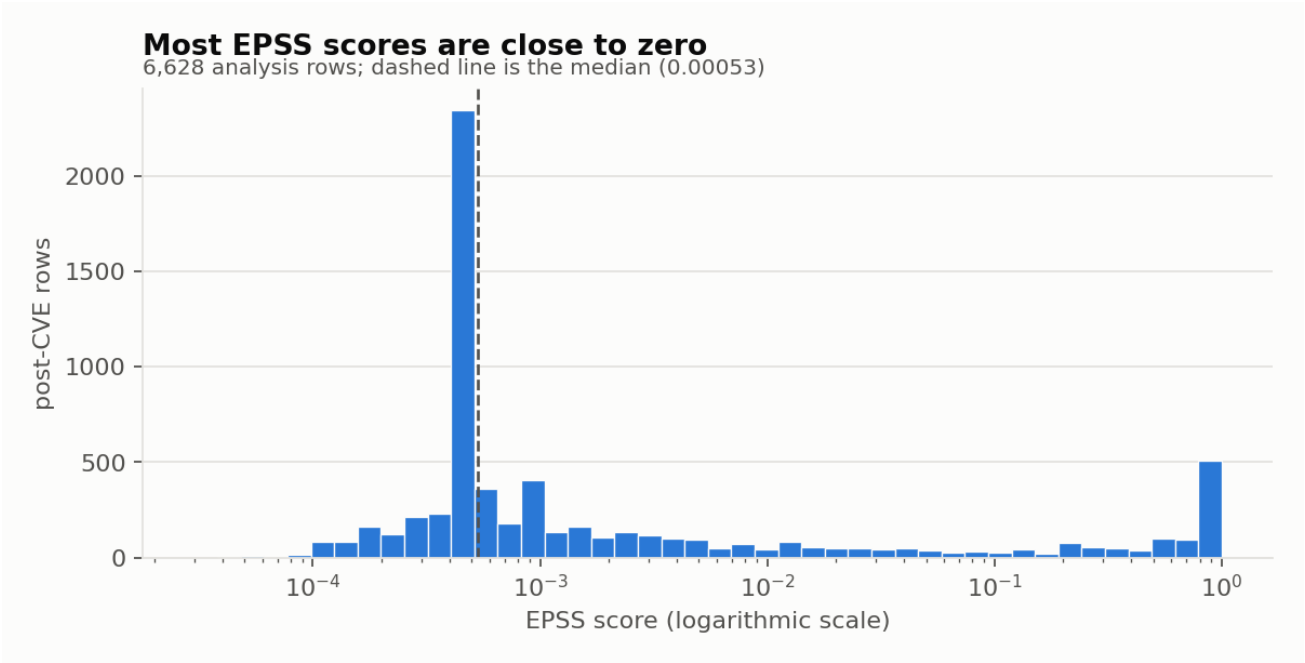


Figure 1. Distribution of EPSS scores. This figure shows how common low and high EPSS scores are. The horizontal axis is EPSS: values near 0 mean very low predicted exploitation probability, while values near 1 mean very high probability. The scale is compressed so very small scores remain visible. The vertical axis counts post-CVE rows, where one row is one link between a collected post and a CVE. The dashed line is the middle score. Most rows have very low EPSS, while a smaller group has much higher scores.

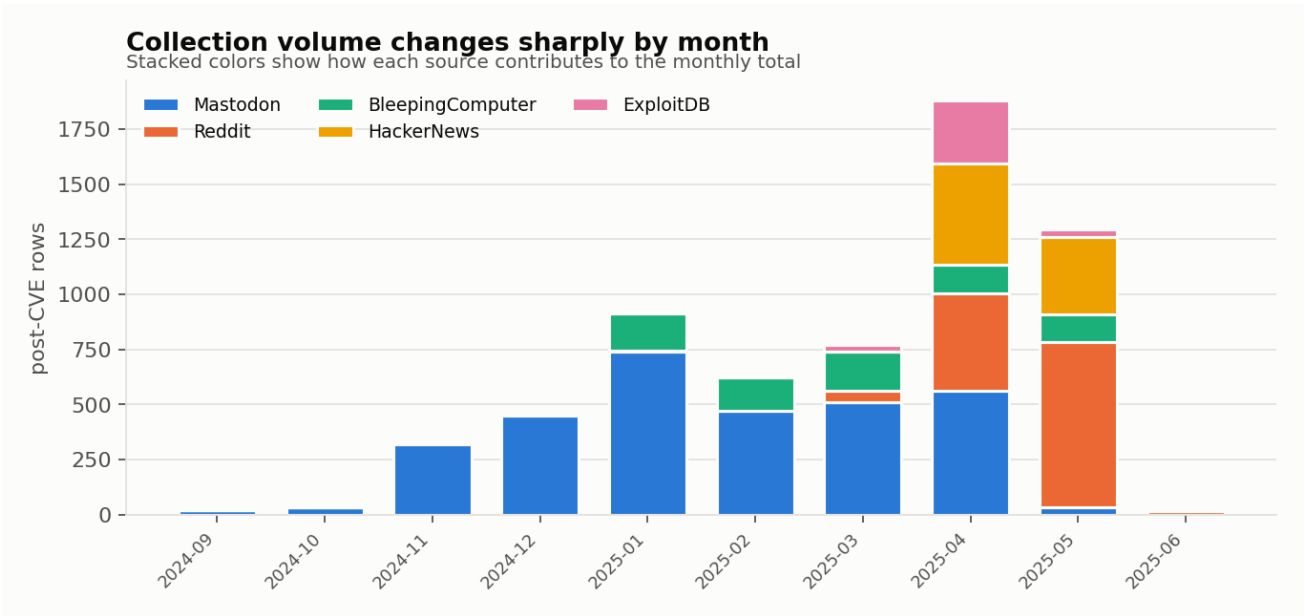


Figure 2. Monthly post volume by source. Each bar represents one month. Its full height is the number of post-to-CVE links collected in that month, and each color shows how many came from one source. A taller bar means more records were collected. Changes in the colors show that the mix of sources also changed over time, so differences between months may partly reflect collection coverage.

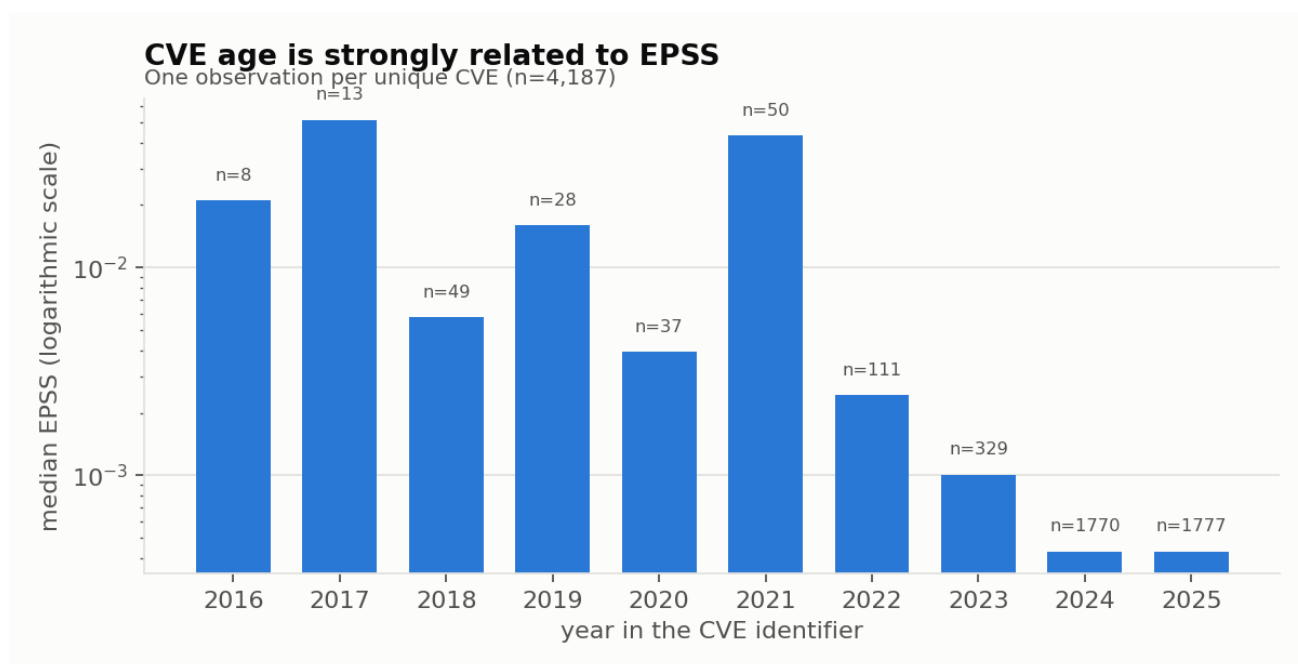


Figure 3. EPSS by CVE year. The horizontal axis is the year written in the CVE identifier, not the year of the social-media post. The vertical axis shows the middle EPSS score for each CVE-year group. The scale is compressed so small values remain visible. Each n label is the number of unique CVEs in that group. Older CVE groups often have higher EPSS, which means CVE age must be considered when groups are compared.

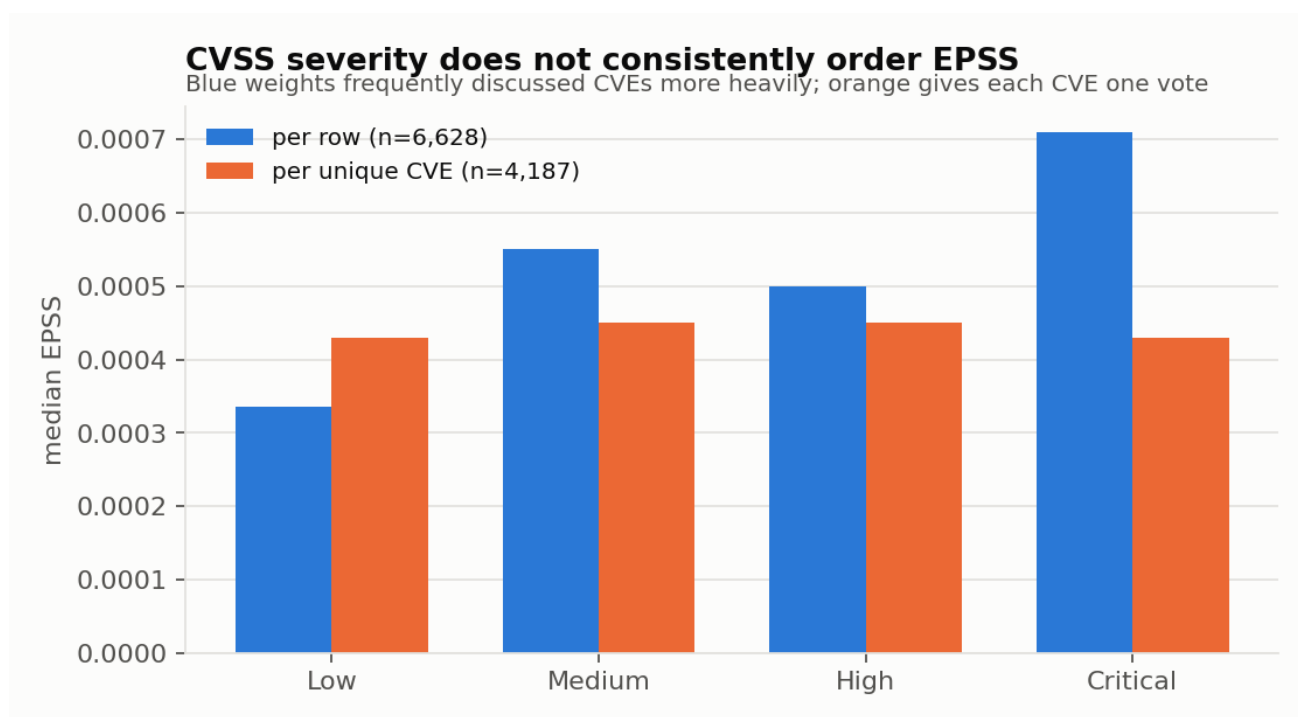


Figure 4. CVSS severity compared with EPSS. The groups along the bottom are the CVSS severity levels Low, Medium, High, and Critical. Bar height is the middle EPSS score. Blue bars count every post-to-CVE link, so a CVE mentioned many times can influence the result repeatedly. Orange bars count each CVE once. The different patterns show that CVSS severity and predicted exploitation probability are related but are not the same thing.

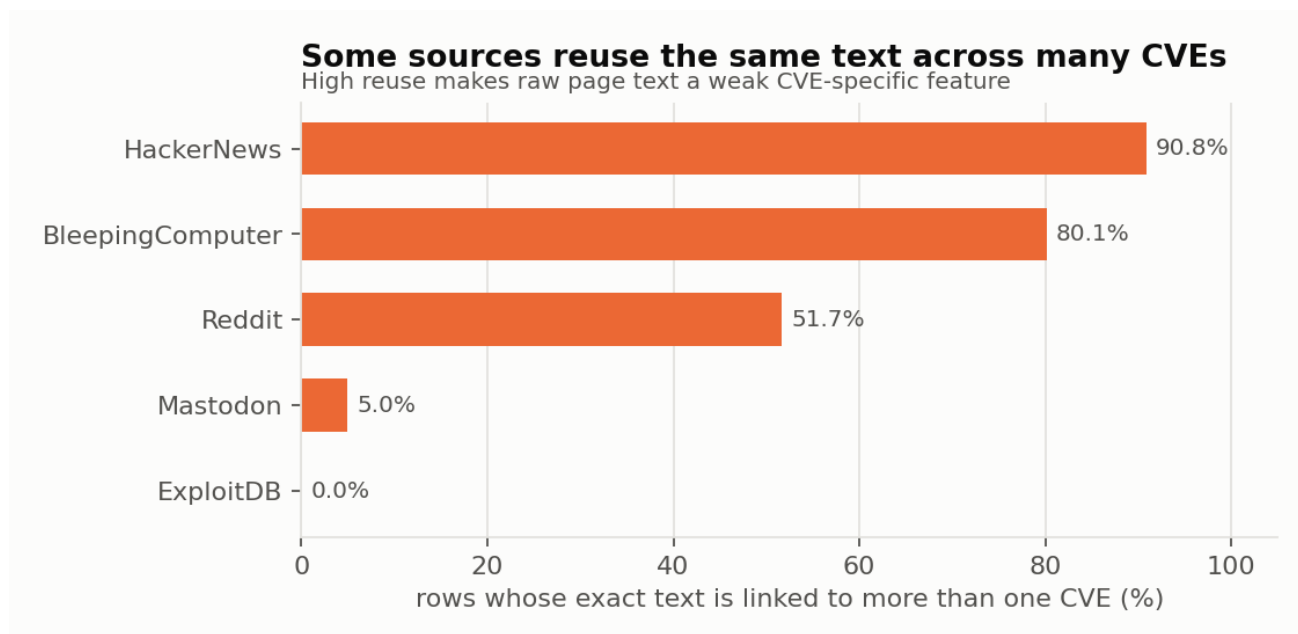


Figure 5. Reuse of post text across CVEs. Each bar represents one source. Bar length is the percentage of its rows where the exact same post text was connected to more than one CVE. Such text is not CVE-specific because it may be a general article or copied page about several vulnerabilities. A high percentage means a text model could learn repeated page content instead of information about one particular CVE.

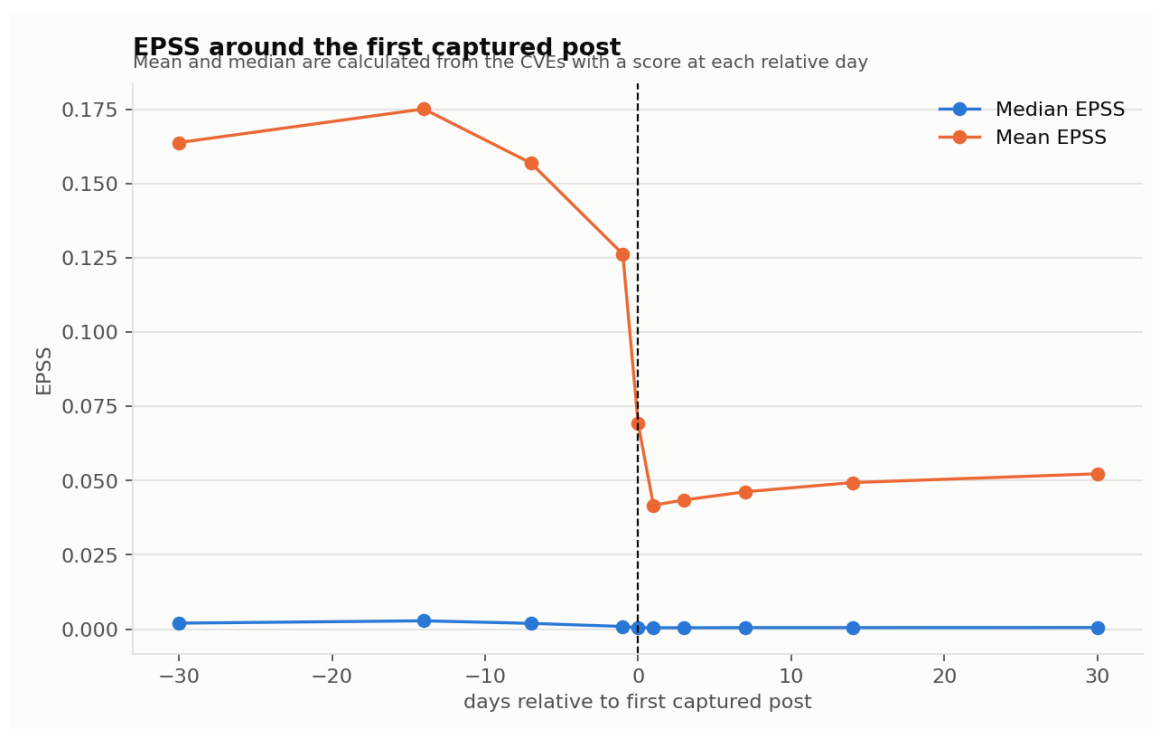


Figure 6. EPSS around the first captured post. Day 0 is the earliest post for a CVE captured in this dataset. Negative day numbers are before that captured post and positive numbers are after it. The blue line is the middle EPSS score, and the orange line is the average among CVEs with data on that day. The average is much higher than the middle value because a small number of high-EPSS CVEs pull it upward.

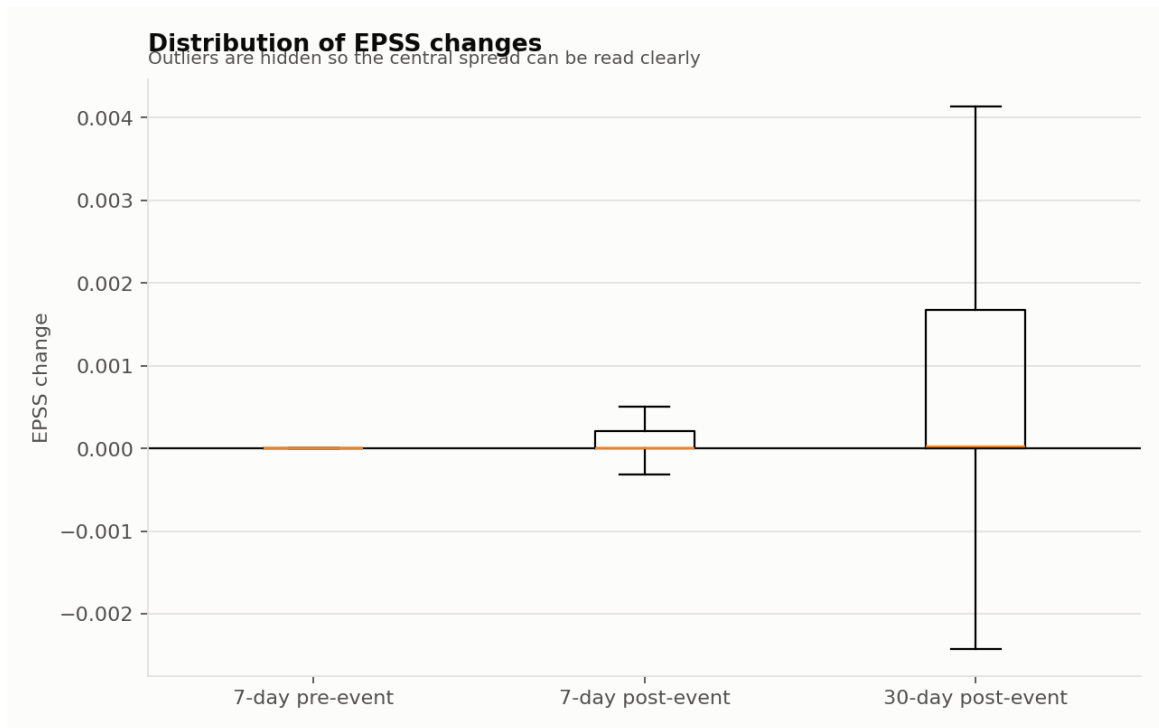


Figure 7. Distribution of EPSS changes. This figure shows how much EPSS changed in several time windows. A value above 0 means EPSS increased; a value below 0 means it decreased. The line inside each box is the middle change, and the box contains the middle half of CVEs. The longer lines show the wider range. Most changes are small, but a smaller number of CVEs change much more.

## 7. Data Quality

| Scope               | Field or field group         | Rows checked | Rows with missing value | Missing (%) | Meaning                                                                                       |
|---------------------|------------------------------|--------------|-------------------------|-------------|-----------------------------------------------------------------------------------------------|
| Each model file     | social_media_post            | 9218         | 5                       | 0.05        | Five absent posts occur outside the analysis population; all analyzed rows contain post text. |
| Each model file     | days_since_latest_git_source | 9218         | 7323                    | 79.44       | Present only when dated Git evidence is available.                                            |
| Each model file     | days_since_oldest_git_source | 9218         | 7323                    | 79.44       | Present only when dated Git evidence is available.                                            |
| Each model file     | source_links                 | 9218         | 1582                    | 17.16       | Absence exactly agrees with sources_available=False.                                          |
| Each model file     | github_urls                  | 9218         | 6879                    | 74.63       | Optional evidence; most rows do not contain a GitHub URL.                                     |
| Analysis population | 21 required core fields      | 6628         | 0                       | 0           | No analyzed row is missing its CVE, date, post text, EPSS, CVSS, or core metadata.            |

Rows checked is the number examined; rows with missing value is how many contain a blank in that field. Some blanks are expected when no external link or GitHub evidence exists. The final row checks all 21 required fields together. All 6,628 analyzed rows contain their CVE, date, original post text, EPSS, CVSS, and other required metadata.

The analysis contains 387 exact texts linked to more than one CVE. They affect 2,191 rows, or 33.1% of the analysis population. HackerNews often contains full-page captures rather than individual comments. A text model can therefore learn page templates instead of CVE-specific evidence.

CVE age is also important because older vulnerabilities have had more time to accumulate exploitation evidence. CVSS and EPSS must remain separate: CVSS describes technical impact, while EPSS estimates exploitation likelihood.

## 8. LLM Summary Variants

| Model   | Summary field           | Available summaries | Missing (%) | Median words | 90th-percentile words | Maximum words |
|---------|-------------------------|---------------------|-------------|--------------|-----------------------|---------------|
| gemma   | External-source summary | 7636                | 17.16       | 81           | 136.5                 | 343           |
| gpt     | External-source summary | 7606                | 17.49       | 111          | 180                   | 561           |
| mistral | External-source summary | 7317                | 20.62       | 122          | 205                   | 504           |
| gemma   | CVSS explanation        | 9218                | 0           | 66           | 82                    | 117           |
| gpt     | CVSS explanation        | 9218                | 0           | 85           | 136                   | 249           |
| mistral | CVSS explanation        | 9016                | 2.19        | 108          | 150                   | 259           |
| gemma   | GitHub summary          | 2381                | 74.17       | 81           | 118                   | 424           |
| gpt     | GitHub summary          | 2246                | 75.63       | 210          | 298.5                 | 1208          |
| mistral | GitHub summary          | 2388                | 74.09       | 113          | 190                   | 830           |

Available summaries counts outputs that are not blank. Missing percentage uses all 9,218 rows. Median words is the middle summary length. The 90th-percentile value means 90% of summaries are no longer than that value. Maximum words is the longest summary.

| Model   | External-source summary gaps | GitHub summary gaps | CVSS explanation gaps |
|---------|------------------------------|---------------------|-----------------------|
| Gemma   | 0                            | 7                   | 0                     |
| GPT     | 30                           | 141                 | 0                     |
| Mistral | 319                          | 0                   | 202                   |

A gap means the LLM received the needed input but no summary was saved. For example, a GitHub gap means a GitHub URL existed but its summary was blank. This separates expected blanks caused by missing source material from possible model or processing failures. It checks availability, not factual accuracy.

Gemma had the best coverage and parsed CVSS agreement. GPT was generally accurate but had more truncated source summaries. Mistral was more verbose and structured but had more missing outputs and more false claims that a recent CVE was not real. The primary timing study therefore uses original dates and historical EPSS rather than LLM summaries.

## 9. EPSS by CVE Year

| CVE year | Unique CVEs | Mean EPSS | Median EPSS |
|----------|-------------|-----------|-------------|
| 2008     | 1           | 0.04015   | 0.04015     |
| 2009     | 1           | 0.93821   | 0.93821     |
| 2010     | 2           | 0.70186   | 0.70186     |
| 2012     | 1           | 0.0138    | 0.0138      |
| 2013     | 3           | 0.54307   | 0.61542     |
| 2014     | 3           | 0.8851    | 0.94237     |
| 2015     | 4           | 0.011168  | 0.007235    |
| 2016     | 8           | 0.21509   | 0.021025    |
| 2017     | 13          | 0.3966    | 0.05142     |
| 2018     | 49          | 0.17916   | 0.00578     |
| 2019     | 28          | 0.25618   | 0.015925    |
| 2020     | 37          | 0.23817   | 0.00392     |
| 2021     | 50          | 0.35679   | 0.04318     |
| 2022     | 111         | 0.13996   | 0.00244     |
| 2023     | 329         | 0.11496   | 0.00101     |
| 2024     | 1770        | 0.028388  | 0.00043     |
| 2025     | 1777        | 0.0019859 | 0.00043     |

CVE year is the year inside the identifier, not the year of the social-media post. Each vulnerability is counted once. Mean EPSS is the average for that group and median EPSS is the middle score. Results based on only a few CVEs are less dependable.

## 10. Monthly Collection Pattern

| Month   | BleepingComputer rows | ExploitDB rows | HackerNews rows | Mastodon rows | Reddit rows |
|---------|-----------------------|----------------|-----------------|---------------|-------------|
| 2021-11 | 0                     | 0              | 0               | 0             | 1           |
| 2021-12 | 0                     | 0              | 0               | 0             | 6           |
| 2022-01 | 0                     | 0              | 0               | 0             | 2           |
| 2022-11 | 0                     | 0              | 0               | 0             | 2           |
| 2023-01 | 0                     | 0              | 0               | 0             | 6           |
| 2023-02 | 0                     | 0              | 0               | 0             | 6           |
| 2023-03 | 0                     | 0              | 0               | 0             | 5           |
| 2023-04 | 0                     | 0              | 0               | 0             | 5           |
| 2023-05 | 0                     | 41             | 0               | 0             | 13          |
| 2023-06 | 0                     | 23             | 0               | 0             | 0           |
| 2023-07 | 0                     | 30             | 0               | 0             | 3           |
| 2023-08 | 0                     | 25             | 0               | 0             | 0           |
| 2023-09 | 0                     | 11             | 0               | 0             | 0           |
| 2023-10 | 0                     | 8              | 0               | 0             | 0           |
| 2023-11 | 0                     | 0              | 0               | 0             | 1           |
| 2024-02 | 0                     | 10             | 0               | 0             | 0           |
| 2024-03 | 0                     | 42             | 0               | 0             | 2           |
| 2024-04 | 0                     | 36             | 0               | 0             | 3           |
| 2024-05 | 0                     | 8              | 0               | 0             | 0           |
| 2024-06 | 0                     | 4              | 0               | 0             | 2           |
| 2024-07 | 0                     | 0              | 0               | 0             | 1           |
| 2024-08 | 0                     | 2              | 0               | 0             | 1           |
| 2024-09 | 0                     | 0              | 0               | 21            | 1           |
| 2024-10 | 0                     | 2              | 0               | 32            | 1           |
| 2024-11 | 0                     | 0              | 0               | 320           | 5           |
| 2024-12 | 0                     | 0              | 0               | 450           | 5           |
| 2025-01 | 168                   | 0              | 0               | 741           | 3           |
| 2025-02 | 154                   | 0              | 0               | 469           | 1           |
| 2025-03 | 178                   | 30             | 0               | 510           | 51          |
| 2025-04 | 130                   | 284            | 462             | 560           | 443         |
| 2025-05 | 125                   | 32             | 354             | 33            | 750         |
| 2025-06 | 0                     | 0              | 0               | 0             | 14          |

Month is shown as year-month. Each number counts post-to-CVE links from that source during that month. Zero means no such row was collected. Because the source mix changes over time, differences between months may reflect collection coverage as well as real activity.

## 11. Why the Analysis Uses Time and Similar CVEs

A simple correlation would mix original post information, LLM prior knowledge, CVE age, technical severity, source coverage, repeated records, and signals that may already influence EPSS. The analysis first verified the data, then examined features and LLM quality, aligned historical EPSS around first-post events, audited pre-event coverage, compared posted CVEs with selected similar CVEs, and finally added KEV as an external outcome.

## 12. Temporal Event Study

For each canonical CVE, the earliest post date captured in this dataset is  $t$ . EPSS was examined 30, 14, 7, and 1 day before  $t$  and 1, 3, 7, 14, and 30 days after it when data existed. Negative offsets are before the event; positive offsets are after it.

The baseline anchor is the starting value. It uses EPSS on the post date when available, otherwise day 1 or day 3. A change, also called a delta, is the later value minus this earlier value. A positive result means EPSS rose; a negative result means it fell.

## 13. Feasibility Audit

| Event window | Has required pre-history | Also has CVE-specific text | Also stays within one EPSS version |
|--------------|--------------------------|----------------------------|------------------------------------|
| +/-7d        | 852                      | 384                        | 372                                |
| +/-14d       | 810                      | 346                        | 321                                |
| +/-30d       | 773                      | 321                        | 213                                |

This table checks whether a fair before-and-after comparison is possible. An event window of plus/minus 7 days means seven days before and seven days after the first post. Required pre-history means EPSS already existed far enough before the post. CVE-specific text keeps first-post text connected to only one CVE and removes general articles or copied pages linked to several CVEs. One EPSS version means the window does not cross a change in the EPSS calculation method. Each column is a further filter, so its count is a subset of the count to its left.

For the plus/minus 30-day window, 773 CVEs had enough earlier EPSS data. Keeping only CVE-specific text reduced this to 321. Requiring one EPSS version reduced it to 213.

The audit's broader text rule retained 3,523 forward events, while the implemented clean-first-post rule retained 3,507. They are different filters. The small before-and-after subset is older and higher-risk, so it cannot represent the whole corpus.

## 14. Temporal Results

### Understanding the two p-values

The t-test p-value asks whether the average change across CVEs is different from zero. It uses the actual numerical changes, so a few unusually large values can strongly affect it.

The Wilcoxon p-value asks whether the non-zero changes generally point in one direction. It ranks changes by size and is less affected by extreme CVEs. Exact zero changes are excluded from this Wilcoxon calculation.

A small p-value, commonly below 0.05, means the pattern would be unusual if there were no consistent change. It does not show that the change is large or useful, and it does not prove that the post caused it. If the two tests disagree, extreme values, many zeros, or an uneven distribution may be influencing the result.

Example: for changes of +0.10, +0.02, -0.01, and 0.00, the t-test uses the average. Wilcoxon removes the zero, ranks the other three changes by size, and compares the positive ranks with the negative rank. The report applies these tests to hundreds or thousands of CVEs.

| EPSS change definition          | CVEs | Mean change | Median change | 25th percentile | 75th percentile | Increased | Decreased | No change | Increased (%) | Decreased (%) | t-test p-value | Wilcoxon p-value |
|---------------------------------|------|-------------|---------------|-----------------|-----------------|-----------|-----------|-----------|---------------|---------------|----------------|------------------|
| Change from t-7 to t-1          | 924  | 0.0059892   | 0             | 0               | 0               | 198       | 83        | 643       | 21.43         | 8.98          | 0.025298       | 3.1442e-06       |
| Change from t-7 to event anchor | 935  | 0.010968    | 0             | 0               | 1e-05           | 235       | 89        | 611       | 25.13         | 9.52          | 0.00055532     | 1.7022e-09       |
| Change from anchor to t+7       | 4092 | 0.0066415   | 0             | 0               | 0.0001          | 1492      | 351       | 2249      | 36.46         | 8.58          | 8.76e-12       | 1.4583e-123      |

| EPSS change definition                     | CVEs | Mean change | Median change | 25th percentile | 75th percentile | Increased | Decreased | No change | Increased (%) | Decreased (%) | t-test p-value | Wilcoxon p-value |
|--------------------------------------------|------|-------------|---------------|-----------------|-----------------|-----------|-----------|-----------|---------------|---------------|----------------|------------------|
| Change from anchor to t+30                 | 4088 | 0.013006    | 0             | 0               | 0.00027         | 2033      | 550       | 1505      | 49.73         | 13.45         | 6.4286e-20     | 4.6796e-154      |
| 7-day post change minus 7-day pre-change   | 916  | 0.0057785   | 0             | -5e-05          | 0.0005725       | 302       | 266       | 348       | 32.97         | 29.04         | 0.13543        | 2.7067e-05       |
| 30-day post change minus 30-day pre-change | 778  | -0.0096739  | 2.5e-05       | -0.001355       | 0.0022675       | 396       | 292       | 90        | 50.9          | 37.53         | 0.16756        | 0.027486         |

t is the first post date, so t-7 is seven days before it and t+30 is 30 days after it. Anchor is the starting EPSS value on day 0, day 1, or day 3. CVEs is the number with both values needed for the comparison. Positive changes are increases and negative changes are decreases. Direction columns count rises, falls, and no change. The two p-values test whether the pattern is likely to be more than random variation, but they do not prove that the post caused it.

Many CVEs rose after the event, but many were unchanged and a smaller moving group influenced the average. Limited pre-event coverage prevents a broad reverse-causality conclusion.

## 15. Forward Comparison with Similar CVEs

Forward means that we start at the posted CVE's first captured post and look 7 or 30 days ahead. We compare its EPSS movement with selected similar CVEs. Statistical studies call this selection process matching, but this report uses the simpler term comparison CVE.

### How the comparison CVEs were selected

Both the posted CVE and every comparison CVE come from the original social-media dataset. For a posted CVE with event date  $t$ , we look for other CVEs whose earliest captured posts occur after  $t+30$ . Those CVEs form the comparison group during the study window. Their later post dates only confirm that no post for them was captured in this dataset from  $t$  through  $t+30$ .

Selection was repeated separately for every posted CVE:

- Start with the posted CVE and its first captured post date.
- Exclude the same CVE and candidates whose own first captured post was not more than 30 days later. This keeps the candidate without a captured post in this dataset throughout the outcome window.
- Keep the same CVE year. Prefer the same CVSS severity; if fewer than 12 candidates exist, allow other severities from the same year.
- Rank candidates by the CVSS-score difference, adding 0.25 for a severity mismatch and 0.05 for a source mismatch. Retain the closest 12 at this stage.
- Compare starting EPSS percentiles on day 0, or day 1 and then day 3 when needed. Remove candidates more than 10 percentile points away.
- Rank the remaining candidates by starting-percentile distance and then technical/source distance. Select up to three comparison CVEs.

A comparison CVE can be reused for different posted CVEs. No captured post during this window refers only to this dataset and does not prove absence from every social platform.

| Coverage measure                             | Value |
|----------------------------------------------|-------|
| CVEs with a baseline EPSS anchor             | 4116  |
| Posted-CVE/comparison-CVE pair rows          | 1668  |
| Posted CVEs with at least one comparison CVE | 884   |

One posted CVE can have several comparison CVEs, so there are more pair rows than posted CVEs. Similarity selection reduces measured differences but does not make different vulnerabilities identical. The comparison shows association and does not prove causation.

### Real comparison CVE examples from the dataset

These examples were checked against the original GPT dataset, the saved event and pair tables, and cached FIRST daily EPSS files. Example A has a positive difference from similar CVEs; Example B has a negative difference. This shows that posted CVEs do not always rise more.

| Example | Role         | CVE            | First captured post | Days after event | Source           | Year / severity / CVSS | Starting EPSS percentile |
|---------|--------------|----------------|---------------------|------------------|------------------|------------------------|--------------------------|
| A       | Posted CVE   | CVE-2024-42733 | 2025-03-10          | 0                | Mastodon         | 2024 / Critical / 9.8  | 0.1187                   |
| A       | Comparison 1 | CVE-2024-48573 | 2025-04-10          | 31               | ExploitDB        | 2024 / Critical / 9.8  | 0.1187                   |
| A       | Comparison 2 | CVE-2024-35056 | 2025-04-11          | 32               | HackerNews       | 2024 / Critical / 9.8  | 0.1187                   |
| A       | Comparison 3 | CVE-2024-50672 | 2025-04-15          | 36               | ExploitDB        | 2024 / Critical / 9.8  | 0.1187                   |
| B       | Posted CVE   | CVE-2023-28461 | 2025-01-09          | 0                | BleepingComputer | 2023 / Critical / 9.8  | 0.97173                  |
| B       | Comparison 1 | CVE-2023-34048 | 2025-03-04          | 54               | BleepingComputer | 2023 / Critical / 9.8  | 0.97409                  |
| B       | Comparison 2 | CVE-2023-26602 | 2025-04-16          | 97               | ExploitDB        | 2023 / Critical / 9.8  | 0.97588                  |
| B       | Comparison 3 | CVE-2023-30258 | 2025-04-11          | 92               | ExploitDB        | 2023 / Critical / 9.8  | 0.98104                  |

Posted CVE is the vulnerability whose first captured post defines the event. Comparison 1 to 3 are the selected similar CVEs. Days after event measures when each comparison CVE's own first captured post occurred; every value is more than 30 days. Year, severity, and CVSS describe the selection profile. Starting EPSS percentile is measured on the posted CVE's event date. A percentile of 0.11870 means a rank above about 11.87% of scored CVEs.

Example A uses CVE-2024-42733, whose first captured post was on 10 March 2025, and comparison CVEs that first appeared 31, 32, and 36 days later. All four are 2024 Critical CVEs with CVSS 9.8 and starting percentile 0.11870. Example B uses CVE-2023-28461 and three 2023 Critical, CVSS 9.8 comparison CVEs. Their starting percentiles are within 0.00931, or 0.931 percentile points, and their posts occurred 54, 92, and 97 days later. The complete CVSS submetrics also happen to agree in these examples, although that is not a general required rule.

| Example | Posted CVE     | Horizon (days) | Posted CVE change | Average comparison-CVE change | Difference from similar CVEs |
|---------|----------------|----------------|-------------------|-------------------------------|------------------------------|
| A       | CVE-2024-42733 | 7              | 0.39616           | 0.22199                       | 0.17417                      |
| A       | CVE-2024-42733 | 30             | 0.41281           | 0.27477                       | 0.13804                      |
| B       | CVE-2023-28461 | 7              | 0.00013           | 0.0017533                     | -0.0016233                   |
| B       | CVE-2023-28461 | 30             | 0.00033           | 0.00187                       | -0.00154                     |

Posted CVE change is its later percentile minus its starting percentile. Average comparison-CVE change is the average of the same calculation for the three similar CVEs over the same dates. Difference from similar CVEs is the posted-CVE change minus that average. Positive means the posted CVE rose more; negative means the comparison CVEs rose more on average.

At seven days in Example A, the posted CVE changed by +0.39616 and the comparison CVEs by +0.22199 on average, giving +0.17417 or 17.417 percentile points. In Example B, the posted CVE changed by +0.00013 and the comparison CVEs by +0.00175 on average, giving -0.00162. These examples explain the calculation; the study conclusion uses all 869 posted CVEs with valid seven-day comparison outcomes.

### How to read the two comparisons

Change in posted CVEs alone measures how each posted CVE's EPSS percentile changed after its first captured social-media post. It answers: did the posted CVE's EPSS percentile increase or decrease after the captured post?

Example: a posted CVE starts at percentile 0.60 and reaches 0.70 after seven days. Its change is 0.70 minus 0.60, which equals +0.10 or 10 percentile points. This shows an increase but does not show whether it was unusual because similar CVEs may also have risen.

Change compared with similar CVEs subtracts the average change of the selected comparison CVEs from the posted CVE's change. It answers: did the posted CVE change more or less than similar CVEs during the same period?

Example: the posted CVE increases by +0.10 and its comparison CVEs increase by an average of +0.03. The difference is +0.10 minus +0.03, which equals +0.07. The posted CVE rose by 7 percentile points more than the similar CVEs. This does not prove that the post caused the increase because both changes may reflect the same developing threat.

### Change in posted CVEs alone

| Horizon (days) | Posted CVEs included | Mean change in posted CVEs | Median change in posted CVEs | Posted CVEs increased (%) | Posted CVEs decreased (%) | Posted CVEs unchanged (%) |
|----------------|----------------------|----------------------------|------------------------------|---------------------------|---------------------------|---------------------------|
| 7              | 4092                 | 0.04299                    | 0.00125                      | 81.92                     | 17.38                     | 0.71                      |
| 30             | 4088                 | 0.060727                   | 0.00612                      | 83.93                     | 15.85                     | 0.22                      |

Horizon is the number of days after the first post. Posted CVEs included is the number with both a starting EPSS percentile and a later percentile at that horizon. Mean change is the average change, and median change is the middle change. Each change is the later percentile minus the starting percentile. Positive values mean upward movement relative to other CVEs, while negative values mean downward movement. The final columns show how often each direction occurred.

### Change compared with similar CVEs

All posted CVEs with selected comparison CVEs:

| Horizon (days) | Posted CVEs compared | Mean difference from similar CVEs | Median difference from similar CVEs | 25th percentile | 75th percentile | Posted CVE rose more (%) | Posted CVE rose less (%) | t-test p-value | Wilcoxon p-value |
|----------------|----------------------|-----------------------------------|-------------------------------------|-----------------|-----------------|--------------------------|--------------------------|----------------|------------------|
| 7              | 869                  | 0.045343                          | 4.6667e-05                          | -0.00033        | 0.0321          | 53.51                    | 43.84                    | 4.9826e-13     | 4.0691e-09       |
| 30             | 868                  | 0.0017783                         | 2e-05                               | -0.031222       | 0.04571         | 50.58                    | 48.04                    | 0.83248        | 0.4811           |

More strictly cleaned first-post group:

| Horizon (days) | Posted CVEs compared | Mean difference from similar CVEs | Median difference from similar CVEs | Posted CVE rose more (%) | Posted CVE rose less (%) |
|----------------|----------------------|-----------------------------------|-------------------------------------|--------------------------|--------------------------|
| 7              | 642                  | 0.054634                          | 0.0001075                           | 56.386                   | 41.121                   |
| 30             | 641                  | -0.0025662                        | 9e-05                               | 51.014                   | 48.05                    |

Posted CVEs compared is the number with the scores and similar CVEs needed for the comparison. For each posted CVE, we subtract the comparison CVEs' average EPSS-percentile change from the posted CVE's change. Mean difference from similar CVEs is the average of these differences, and median difference is the middle one. A positive result means the posted CVE rose more than its comparison CVEs; a negative result means it rose less or fell more. The direction percentages show how often each result occurred. They may not add to 100% because some posted CVEs changed by the same amount as their comparison CVEs. The more strictly cleaned group keeps first-post text connected to only one CVE and removes general or copied text associated with several CVEs. The 25th and 75th percentiles contain the middle half of the differences. P-values test whether the differences consistently move away from zero; they do not prove causation.

At seven days, the mean difference from similar CVEs was positive, but the median was only 0.000047 and posted CVEs rose more than their comparison CVEs in 53.51% of comparisons. The more strictly cleaned group reached 56.39%. At 30 days, the tests were not significant and the mean for the more strictly cleaned group was slightly negative. This supports a small, short-term association, not a causal effect.

## 16. CISA KEV Outcome

| KEV measure                        | CVE count | Share of events (%) |
|------------------------------------|-----------|---------------------|
| All CVE events                     | 4187      | 100                 |
| Ever listed in CISA KEV            | 284       | 6.78                |
| Added to KEV on or after the event | 119       | 2.84                |
| Added to KEV within 90 days        | 102       | 2.44                |

CISA KEV is the US Cybersecurity and Infrastructure Security Agency's catalog of vulnerabilities with evidence of real-world exploitation. It is used as an external confirmation of known exploitation, not as another EPSS score. The rows are smaller groups taken from the 4,187 first-post events:

- All CVE events: all 4,187 unique CVE events in the study.
- Ever listed in CISA KEV: 284 CVEs found in the downloaded catalog, whether added before or after the first post.
- Added on or after the event: 119 CVEs whose catalog date was the same as or later than the first post.
- Added within 90 days: 102 CVEs added from the event date through day 90. These are part of the 119 above, which are part of the 284 ever listed.

CVE count is the number in each group. Share of events divides that count by 4,187. For example, 5 CVEs out of 100 would be a 5% share. CISA's dateAdded is the catalog-entry date, not necessarily the first exploitation date; exploitation may have started earlier.

## 17. Exploratory Predictive Results

| Prediction target                   | Predictor set                      | CVEs | Positive outcome share | Mean ROC-AUC | ROC-AUC standard deviation | Mean PR-AUC | PR-AUC standard deviation |
|-------------------------------------|------------------------------------|------|------------------------|--------------|----------------------------|-------------|---------------------------|
| EPSS percentile increased by day 7  | CVSS only                          | 4116 | 0.81438                | 0.57703      | 0.018383                   | 0.84318     | 0.010889                  |
| EPSS percentile increased by day 7  | CVSS plus baseline EPSS            | 4116 | 0.81438                | 0.66141      | 0.014423                   | 0.889       | 0.0084572                 |
| EPSS percentile increased by day 7  | Plus source and attention features | 4116 | 0.81438                | 0.71239      | 0.0089329                  | 0.90468     | 0.0058811                 |
| EPSS percentile increased by day 30 | CVSS only                          | 4116 | 0.83358                | 0.60981      | 0.010586                   | 0.87213     | 0.0010864                 |
| EPSS percentile increased by day 30 | CVSS plus baseline EPSS            | 4116 | 0.83358                | 0.71257      | 0.014979                   | 0.92001     | 0.0062109                 |
| EPSS percentile increased by day 30 | Plus source and attention features | 4116 | 0.83358                | 0.74662      | 0.021445                   | 0.9329      | 0.0091467                 |
| Added to CISA KEV within 90 days    | CVSS only                          | 4116 | 0.023567               | 0.62114      | 0.056396                   | 0.041009    | 0.0063279                 |
| Added to CISA KEV within 90 days    | CVSS plus baseline EPSS            | 4116 | 0.023567               | 0.67369      | 0.061344                   | 0.07912     | 0.019098                  |
| Added to CISA KEV within 90 days    | Plus source and attention features | 4116 | 0.023567               | 0.92554      | 0.035879                   | 0.5236      | 0.073085                  |

Prediction target is the result the model tries to predict; predictor set is the information given to it. Positive outcome share is how often the result occurred. ROC-AUC shows how well positive cases are ranked above negative cases: 0.5 is similar to chance and 1.0 is perfect. PR-AUC focuses on correctly finding the uncommon positive cases. Standard deviation shows how much performance changed across repeated validation groups.

The largest AUCs are exploratory. The source-and-attention set includes post totals and source counts calculated across the full corpus, so some information can occur after the event. It also includes `epss_status`, which is related to score collection. Validation folds were random rather than chronological. The 0.9255 KEV ROC-AUC is therefore not proven real-time performance.

## 18. Final Interpretation

The analysis found a small, short-term relationship between social-media attention and changes in EPSS. During the seven days after the first captured post, posted CVEs showed a slightly greater average increase than their selected comparison CVEs. The typical difference was very small, and only a modest majority of posted CVEs increased more than the similar CVEs. After 30 days, there was no clear evidence that the difference continued. This does not prove that social-media posts caused EPSS to rise or caused exploitation. Social-media attention is best treated as a possible early-warning signal alongside EPSS, CVSS, source context, and confirmed exploitation evidence.

## 19. Verification Appendix

| Verification check                                               | Passed | Local source file                               |
|------------------------------------------------------------------|--------|-------------------------------------------------|
| Three input files have 9,218 rows and 28 columns                 | True   | model_inventory.csv                             |
| Independent CSV row counts agree                                 | True   | model_inventory.csv                             |
| Every parsed source record has exactly 28 columns                | True   | source_file_structure_audit.csv                 |
| Static post/CVE fields match across LLM variants                 | True   | static_feature_consistency.csv                  |
| The analysis population has 6,628 rows and valid dates           | True   | gpt_combined_summ.csv                           |
| All 21 core fields are complete in the analysis population       | True   | source_missingness_audit.csv                    |
| The input contains 5,692 canonical CVEs                          | True   | gpt_combined_summ.csv                           |
| The analysis population and first-event table contain 4,187 CVEs | True   | forward_events_base.csv                         |
| Platform and monthly totals reconcile to 6,628 rows              | True   | platform profile and saved monthly volume table |
| CVE-year totals reconcile to 4,187 unique CVEs                   | True   | epss_by_cve_year.csv                            |
| All summary tables used in the report have complete cells        | True   | saved report-input CSV tables                   |
| Shared-text result is 387 texts and 2,191 analysis rows          | True   | gpt_combined_summ.csv                           |
| Forward EPSS summaries match expected event counts               | True   | forward_event_outcome_summary.csv               |

| Verification check                                                                                  | Passed | Local source file                                        |
|-----------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------|
| Comparison summary contains 869 (+7d) and 868 (+30d) posted-CVE events                              | True   | comparison_summary.csv                                   |
| KEV counts are internally consistent                                                                | True   | kev_summary.csv                                          |
| Worked comparison CVEs and captured-post dates agree with the original dataset                      | True   | gpt_combined_summ.csv and forward_events_base.csv        |
| Worked comparison CVEs first appear more than 30 days after their posted-CVE events                 | True   | comparison_pair_outcomes.csv                             |
| Worked comparison CVEs reproduce the year, CVSS, source-distance, and starting-EPSS selection rules | True   | forward_events_base.csv and comparison_pair_outcomes.csv |
| Worked EPSS values agree with cached FIRST daily scores                                             | True   | Forward_EPSS_Study/epss_cache                            |
| Worked posted-minus-comparison calculations reproduce the saved results                             | True   | comparison_pair_outcomes.csv                             |

Each row is a fact calculated again from a saved data file. True means the recalculated value agrees with this report; generation stops if any check is false. All printed tables have complete cells, and the platform, monthly, and CVE-year totals agree. Missing EPSS observations were not invented or filled in; each result reports how many CVEs had the dates it required.

## 20. Reproducibility Record

The PDF was built from the verified inventory, correlation report, temporal feasibility audit, historical EPSS event panel, comparison-study outputs, KEV enrichment, figures, and saved result tables. External EPSS and KEV information came from the websites listed in Section 3. Report generation used cached local copies so later online updates could not silently change these results.